THE NEXT-GEN WITNESS SKETCH STUDIO

ECHO-FACE AI

From a witness description to a traceable forensic-style portrait.

Describe → Generate → Refine → Review → Export

24-HOUR BUILD • GENERATIVE AI • REACT • SECURE BACKEND

The problem: time is critical

01 ARTIST BOTTLENECK

Specialist forensic artists may not be immediately available.

02 MEMORY FADES

Witness recollection can change over time and with repeated questioning.

03 LIMITED TRACEABILITY

Untracked edits make it difficult to reconstruct how a portrait evolved.

Opportunity: faster visual exploration with human review — not automated identification.

Our solution in one sentence

A secure AI sketch studio where investigators translate a witness description into an editable portrait — while preserving every revision and access event.

FOR THE WITNESS

Describe features in plain language.

FOR THE INVESTIGATOR

Review, refine, compare and export.

The 5-step product journey

1 CREATE CASE
Authorized investigator opens a case.

2 DESCRIBE
Enter facial features and uncertain details.

3 GENERATE
AI creates a draft composite portrait.

4 REFINE
“Narrower jaw; keep everything else.”

5 EXPORT
Save sketch + version history + audit trail.

Live demo: the moment judges remember

WITNESS INPUT

“Adult with an oval face, short dark hair, thick eyebrows and a broad nose.”

Refinement: “Make the jaw slightly narrower, but preserve the eyes.”

WHAT JUDGES SEE

Initial portrait → refined portrait

Before/after slider or two-up comparison

Version 1 and Version 2 in case history

Exported case report

Architecture: buildable in 24 hours

REACT + TAILWIND

- Witness prompt form
- Sketch gallery
- Refinement chat
- Case history

FASTAPI

- Auth + case APIs
- Prompt orchestration
- Image versioning
- Export endpoints

AI + STORAGE

- Image-generation API
- SQLite / PostgreSQL
- Private image storage
- Append-only audit records

Generation depends on configured model access, latency and rate limits; use a cached demo fallback.

Why our approach is different

CONTROLLED REFINEMENT

Edit one attribute at a time; keep the previous portrait available.

UNCERTAINTY-AWARE

Mark uncertain traits rather than inventing confidence.

REPRODUCIBLE HISTORY

Save prompts, timestamps, model metadata and versions.

HUMAN-IN-THE-LOOP

Every result is reviewed and approved by an investigator.

Security and responsible use

SECURE BY DESIGN

Role-based case access

Access audit: who / what / when

Hash-linked audit events

Protected exports and storage

IMPORTANT SAFEGUARDS

Composite is an investigative aid, not proof

No identity match or suspect ranking in MVP

No training on witness data

Avoid suggestive prompting; allow “unknown”

Never claim forensic accuracy or tamper-proof storage without independent validation.

24-hour execution plan

00–03h

Scope + UI

Screens, case schema, API contracts

03–09h

Core pipeline

Prompt → image → saved version

09–15h

Refinement

Chat edits, history, compare

15–20h

Trust + export

Auth, audit, downloadable report

20–24h

Test + rehearse

Fix blockers, backup demo, pitch

What we will prove — not just promise

MVP ACCEPTANCE CHECKS

- ✓ New case can be created
- ✓ Prompt generates a portrait
- ✓ Feedback produces a new version
- ✓ Earlier version can be restored
- ✓ Access event appears in audit log

MEASURABLE DEMO

- Record generation latency
- Verify version count after edits
- Verify export includes case ID
- Verify unauthorized access is rejected
- Demonstrate graceful API failure

The ask: judge us on a working workflow

Not just an AI image.

A controlled, auditable human-AI process.

ECHO-FACE AI | Faster iteration. Clearer provenance. Human judgment first.

Thank you • Live demo + questions